

Fundamentals of Quantum Computing: A Lecture Note

Abstract

These notes introduce quantum computing through its foundation.

Keywords: Quantum Computing, Quantum Mechanics, Qiskit, IBM Quantum

1 Postulates of Quantum Mechanics

Quantum theory represents a fundamentally new kind of probability theory. Nature does not obey classical probability rules.

Classical case:

$$P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

for mutually exclusive events A_1, A_2 .

Quantum case:

$$\psi = \psi_1 + \psi_2, \quad P = |\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2|\psi_1||\psi_2|\cos\theta$$

where θ is the relative phase between ψ_1 and ψ_2 .

The additional interference term gives rise to nonclassical behavior such as *superposition* and *entanglement*. Thus, the state space of a quantum system is a *Hilbert space* $\mathcal{H}$.

A single qubit is represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

1.1 Key Questions in Quantum Mechanics

1. How do we specify the state of a quantum system?
2. How does it evolve in time?
3. What happens when we measure it?
4. How do we describe a composite system?

Postulate 1: The State of a Quantum System

Quantum state $|\psi\rangle$ is a unit vector in a complex Hilbert space $\mathcal{H}$.

Postulate 2: Observables

Observables correspond to Hermitian operators on $\mathcal{H}$.

Example: Spin Observables

For a spin- $\frac{1}{2}$ particle, the spin components along the three axes are represented by Hermitian operators:

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each has eigenvalues $\pm\hbar/2$, corresponding to the measurable spin projections along the respective axes. The eigenstates of S_z are $\{|0\rangle, |1\rangle\}$, while those of S_x and S_y are:

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle), \quad |\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle).$$

These states correspond to spin-up and spin-down orientations along the x and y axes respectively.

Postulate 3: Time Evolution

Time evolution is unitary: $|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$.

Schrödinger Equation

Time evolution is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle,$$

where $H(t)$ is the Hamiltonian (a Hermitian operator representing the energy of the system). If H is time-independent, the solution is

$$|\psi(t)\rangle = e^{-iH(t-t_0)/\hbar} |\psi(t_0)\rangle.$$

Since H is Hermitian, $U(t, t_0) = e^{-iH(t-t_0)/\hbar}$ is unitary, ensuring that the norm $\langle\psi(t)|\psi(t)\rangle$ is preserved.

Postulate 4: Measurement

Measurement outcomes correspond to eigenvalues a_i of an observable A , with probability $p(i) = |\langle a_i | \psi \rangle|^2$ and post-measurement state $|a_i\rangle$.

Measurement and the Stern–Gerlach Experiment

Consider an observable A with eigenvalues $\{a_i\}$ and orthonormal eigenstates $\{|a_i\rangle\}$. Measurement of A on a system in state $|\psi\rangle$ yields:

$$p(i) = |\langle a_i | \psi \rangle|^2,$$

and the post-measurement state collapses to $|a_i\rangle$ corresponding to the observed outcome.

Example: Stern–Gerlach experiment. In 1922, Stern and Gerlach passed a beam of silver atoms through an inhomogeneous magnetic field. Each atom, having a single unpaired electron, behaves as a magnetic dipole. Classically one expects a continuous distribution of deflection angles. Experimentally, however, the beam splits into two distinct spots — demonstrating quantization of spin.

If the incoming spin state is

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1,$$

then measurements of S_z yield

$$p(0) = |a|^2, \quad p(1) = |b|^2.$$

Each measurement collapses the state to either $|0\rangle$ (spin-up) or $|1\rangle$ (spin-down).

Sequential spin measurements. If a beam of spin-up ($S_z = +\hbar/2$) particles passes through another Stern–Gerlach apparatus aligned along x , the output again splits into $S_x = \pm\hbar/2$ components — indicating that S_x and S_z cannot be simultaneously definite. Performing a third measurement of S_z after S_x restores both outcomes, showing that the act of measuring one observable disturbs the other.

Non-commuting observables and uncertainty. For spin operators,

$$[S_x, S_z] = i\hbar S_y.$$

Hence, they do not commute, and the corresponding measurements obey the uncertainty relation:

$$\Delta S_x \Delta S_z \geq \frac{\hbar}{2} |\langle S_y \rangle|.$$

This illustrates the general principle that precise knowledge of one observable precludes exact knowledge of another non-commuting one.

Postulate 5: Composite Systems

For multiple systems, $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n$.

Summary of the Postulates

1. **Quantum states:** vectors in a complex Hilbert space.
2. **Observables:** Hermitian operators with real eigenvalues.
3. **Time evolution:** unitary and norm-preserving, governed by Schrödinger dynamics.
4. **Measurement:** probabilistic, leading to state collapse.
5. **Composite systems:** described by tensor products of Hilbert spaces.

These postulates form the mathematical foundation of quantum mechanics and underlie all quantum computation models.

2 Quantum Computing Basics

2.1 Single Qubits and the Bloch Sphere

A single qubit is the simplest nontrivial quantum system. Mathematically, it is represented as a **normalized vector** in a two-dimensional complex Hilbert space:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Here, $|0\rangle$ and $|1\rangle$ form an *orthonormal basis*:

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = 0.$$

The coefficients α and β are known as the **probability amplitudes**. Upon measurement in the computational basis:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2,$$

and the qubit *collapses* to $|0\rangle$ or $|1\rangle$ respectively.

Global phase invariance: Multiplying a state by an overall complex phase $e^{i\gamma}$ does not change any physical prediction:

$$|\psi\rangle \sim e^{i\gamma} |\psi\rangle.$$

Therefore, two qubit states that differ only by a global phase are physically indistinguishable.

Parameterization of a qubit: Any normalized qubit state can be written (up to a global phase) as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle,$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ are real parameters.

- θ — the **polar angle**, representing the latitude on the Bloch sphere,
- ϕ — the **azimuthal angle**, representing rotation around the Z -axis.

The point (θ, ϕ) uniquely identifies a qubit state on the unit sphere known as the **Bloch sphere**.

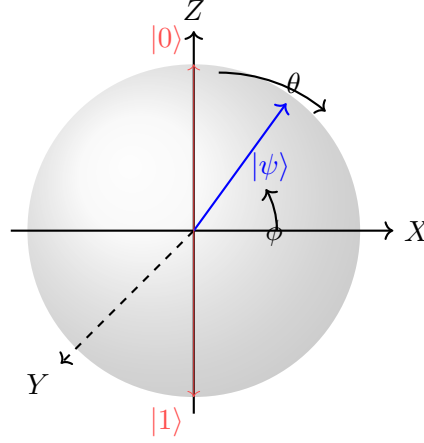
Bloch vector representation: A pure qubit state can equivalently be represented as a point on a unit sphere in $\mathbb{R}^3$ by defining the Bloch vector:

$$\vec{r} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}.$$

Then the corresponding **density matrix** is

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}),$$

where $\vec{\sigma} = (X, Y, Z)$ are the Pauli matrices. This representation provides a convenient geometric intuition of qubit operations as *rotations* on the sphere.



Examples:

- $|0\rangle$ corresponds to $(\theta = 0, \phi = 0)$ — the north pole of the sphere.
- $|1\rangle$ corresponds to $(\theta = \pi, \phi = 0)$ — the south pole.
- The *Hadamard state* $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ lies on the equator at $\phi = 0, \theta = \pi/2$.

Hence, any single-qubit gate can be understood geometrically as a **rotation of the Bloch vector** around a specified axis. This geometric picture becomes extremely useful in visualizing quantum state evolution and gate operations.

2.2 Single-Qubit Gates

Quantum gates are the basic building blocks of quantum circuits. They are represented by **unitary operators** acting on qubits, i.e.,

$$U^\dagger U = U U^\dagger = I.$$

This property ensures that quantum evolution is *reversible* and preserves the total probability ($|\alpha|^2 + |\beta|^2 = 1$).

For a single qubit, any gate can be represented as a 2×2 unitary matrix acting on the vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Pauli Gates: X , Y , and Z

The three Pauli operators form the fundamental set of single-qubit quantum gates:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- **Pauli-X gate (bit flip):** Analogous to the classical NOT gate. It exchanges $|0\rangle$ and $|1\rangle$:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$

On the Bloch sphere, it corresponds to a rotation of π radians about the X -axis:

$$X = R_x(\pi).$$

- **Pauli-Y gate (bit and phase flip):** Combines a bit flip with a relative phase change. It produces a rotation of π about the Y -axis:

$$Y = R_y(\pi).$$

- **Pauli-Z gate (phase flip):** Leaves $|0\rangle$ unchanged but flips the phase of $|1\rangle$:

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Geometrically, it corresponds to a π rotation about the Z -axis:

$$Z = R_z(\pi).$$

These three matrices satisfy the commutation and anticommutation relations:

$$[X, Y] = 2iZ, \quad [Y, Z] = 2iX, \quad [Z, X] = 2iY,$$

and together with the identity form a complete basis for single-qubit operations.

Hadamard Gate (H)

The **Hadamard gate** plays a special role because it creates **superposition**. It transforms the computational basis $\{|0\rangle, |1\rangle\}$ into the *Hadamard basis* $\{|+\rangle, |-\rangle\}$ defined by

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

The matrix form is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its action is given by

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

On the Bloch sphere, it corresponds to a rotation that maps the Z -axis to the X -axis (a π rotation about the diagonal axis $(X + Z)/\sqrt{2}$).

Phase Gates: S and T

Phase gates introduce relative phase shifts between basis states without changing measurement probabilities.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

- The **S gate** (also called the *phase gate*) performs a $\pi/2$ rotation about the Z -axis: $S = R_z(\pi/2)$.
- The **T gate** (also called the $\pi/8$ gate) performs a smaller $\pi/4$ rotation about the Z -axis: $T = R_z(\pi/4)$.

Since any single-qubit unitary can be represented as a combination of rotations about two orthogonal axes, these gates form the foundation of quantum circuit design.

General Single-Qubit Rotation

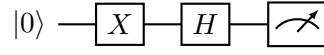
A general rotation on the Bloch sphere can be expressed as:

$$R_{\hat{n}}(\theta) = e^{-i\frac{\theta}{2}(\hat{n}\cdot\vec{\sigma})},$$

where $\hat{n} = (n_x, n_y, n_z)$ is the unit vector indicating the axis of rotation, $\vec{\sigma} = (X, Y, Z)$ is the vector of Pauli matrices, and θ is the rotation angle.

Example Circuit

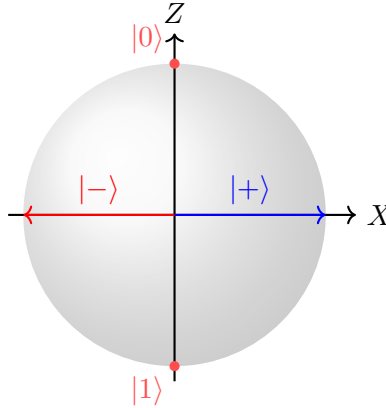
The following example shows a simple circuit that first flips a qubit from $|0\rangle$ to $|1\rangle$ using an X gate and then creates a superposition using a Hadamard gate:



The output after the H gate is

$$HX|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle,$$

which lies on the equator of the Bloch sphere at azimuthal angle $\phi = \pi$.



Thus, the action of single-qubit gates can always be understood geometrically as a rotation of the Bloch vector about some axis in three-dimensional space. This geometric insight is central to quantum control, circuit optimization, and understanding interference phenomena.

2.3 Multiple Qubits and Tensor Products

Two qubits form a 4-dimensional Hilbert space:

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle.$$

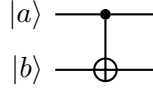
Product state: separable into individual qubits. **Entangled state:** cannot be factorized (e.g., $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$).

2.4 Quantum Circuits and Reversibility

Quantum gates must be unitary, ensuring reversibility.

Example: Controlled-NOT (CNOT) gate.

$$\text{CNOT: } |a, b\rangle \mapsto |a, b \oplus a\rangle.$$



2.5 Universality of Quantum Gates

A **universal gate set** can approximate any unitary transformation:

$$\{H, T, \text{CNOT}\}$$

is universal for quantum computation.

2.6 Multiple Qubits and Tensor Products

Quantum computation becomes truly powerful when we consider systems composed of more than one qubit. If each qubit is described by a Hilbert space $\mathcal{H}_i = \mathbb{C}^2$, then a system of n qubits is described by the **tensor product space**:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n,$$

which has dimension 2^n .

Two-Qubit States

For two qubits, $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$. A general two-qubit state is therefore expressed as

$$|\psi\rangle_{12} = \sum_{i,j \in \{0,1\}} \alpha_{ij} |i\rangle_1 \otimes |j\rangle_2 = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle,$$

with normalization

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1.$$

The combined state space contains all possible linear superpositions of the four computational basis states

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

Product and Entangled States

A two-qubit state is said to be a **product state** (or separable) if it can be written as a tensor product of individual qubit states:

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle.$$

For example:

$$|\psi_1\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\psi_2\rangle = \gamma |0\rangle + \delta |1\rangle \Rightarrow |\psi\rangle_{12} = \alpha\gamma |00\rangle + \alpha\delta |01\rangle + \beta\gamma |10\rangle + \beta\delta |11\rangle.$$

However, many quantum states cannot be expressed in this product form. Such states are called **entangled states**, and they exhibit correlations that have no classical counterpart.

A canonical example is the *Bell state*:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

which satisfies

$$\text{not separable: } \nexists |\psi_1\rangle, |\psi_2\rangle \text{ such that } |\Phi^+\rangle = |\psi_1\rangle \otimes |\psi_2\rangle.$$

Measurement of one qubit in this state immediately determines the outcome of the other, regardless of distance—this is a hallmark of quantum entanglement.

The set of all two-qubit entangled states forms a continuum; the four maximally entangled Bell states are:

$$\begin{aligned} |\Phi^\pm\rangle &= \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \\ |\Psi^\pm\rangle &= \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \end{aligned}$$

These play a central role in quantum communication and quantum algorithms.

Density Matrix Viewpoint

The state of a subsystem in an entangled pair can be described by a **reduced density matrix** obtained via the partial trace. For $|\Phi^+\rangle$,

$$\rho_1 = \text{Tr}_2(|\Phi^+\rangle \langle \Phi^+|) = \frac{1}{2}I,$$

showing that each individual qubit is in a completely mixed state even though the overall system is pure.

2.7 Quantum Circuits and Reversibility

Quantum circuits are composed of **quantum gates**, each represented by a unitary matrix. Because unitaries preserve the inner product, the entire circuit is reversible—no information is lost during computation.

Controlled Operations

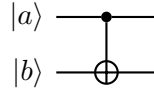
When a gate's action on one qubit depends on the state of another, we obtain a **controlled gate**. The most common example is the **Controlled-NOT (CNOT)** gate, which acts on two qubits as:

$$\text{CNOT: } |a, b\rangle \mapsto |a, b \oplus a\rangle.$$

Its matrix representation in the computational basis is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The CNOT flips the target qubit if and only if the control qubit is in the state $|1\rangle$.



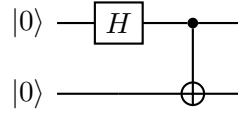
Action examples:

$$\text{CNOT} |00\rangle = |00\rangle, \quad \text{CNOT} |01\rangle = |01\rangle, \quad \text{CNOT} |10\rangle = |11\rangle, \quad \text{CNOT} |11\rangle = |10\rangle.$$

The CNOT gate is a fundamental two-qubit entangling gate. For example:

$$\text{CNOT}(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle.$$

Thus, by applying a Hadamard gate on the first qubit followed by a CNOT, one can create a maximally entangled Bell state.



This small circuit is a foundational building block in quantum teleportation, superdense coding, and many quantum algorithms.

Reversibility and Unitarity

Every quantum circuit represents a unitary transformation U . The inverse circuit corresponds to the Hermitian adjoint $U^\dagger$, which can be implemented by reversing the order of gates and replacing each gate with its inverse. This ensures that information can always be recovered, unlike classical irreversible logic (e.g., AND, OR).

Reversible computation: Quantum mechanics enforces that time evolution be reversible. Hence, every valid quantum operation is represented by a unitary matrix, and measurement (which is non-unitary) marks the end of computation.

2.8 Universality of Quantum Gates

A set of gates is said to be **universal** if any unitary transformation on an arbitrary number of qubits can be approximated to arbitrary precision using only those gates.

In classical computation, $\{\text{AND}, \text{NOT}\}$ or $\{\text{NAND}\}$ is universal for Boolean logic.

In quantum computation, a commonly used universal gate set is:

$$\{H, T, \text{CNOT}\}.$$

- The **Hadamard gate** introduces superposition.
- The **T gate** introduces an irrational phase, necessary for generating all possible rotations.
- The **CNOT gate** introduces entanglement between qubits.

Together, these gates can approximate any unitary operation on n qubits with arbitrary accuracy, as guaranteed by the *Solovay–Kitaev theorem*. This property is crucial for constructing any quantum algorithm.

Universal Gate Concept

Single-qubit gates (H , T)
+ Two-qubit gate (CNOT)

$\Downarrow$

Any Unitary Operation

Thus, universality establishes that quantum circuits composed only of simple building blocks like H , T , and CNOT can, in principle, perform any computation allowed by quantum mechanics.

3 Entanglement, Bell States, and Quantum Teleportation

3.1 The Nature of Entanglement

Entanglement is one of the defining features of quantum mechanics. It occurs when the state of a composite system cannot be expressed as a tensor product of its subsystems' individual states. That is, for a two-qubit state $|\psi\rangle_{12}$,

$$|\psi\rangle_{12} \neq |\psi_1\rangle \otimes |\psi_2\rangle.$$

When this happens, measurement outcomes on one qubit are perfectly correlated (or anti-correlated) with outcomes on the other, even when separated by large distances. This “nonlocal” correlation defies classical intuition and is what Einstein called “spooky action at a distance.”

Example. Consider the two-qubit state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

If we measure the first qubit and obtain 0, the second qubit collapses immediately to $|0\rangle$. If we obtain 1, the second qubit collapses to $|1\rangle$. Thus, both results are perfectly correlated.

The key property of entanglement is that the reduced state of each qubit alone is completely mixed:

$$\rho_1 = \text{Tr}_2(|\Phi^+\rangle \langle \Phi^+|) = \frac{1}{2}I.$$

This means that although the total system is in a pure state, each subsystem individually appears maximally uncertain.

3.2 The Bell States and Their Properties

The four maximally entangled two-qubit states, called the **Bell states**, form an orthonormal basis of the 4-dimensional Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \\ |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \end{aligned}$$

They satisfy the orthonormality and completeness relations:

$$\langle B_i | B_j \rangle = \delta_{ij}, \quad \sum_i |B_i\rangle \langle B_i| = I_{4 \times 4}.$$

Correlation structure.

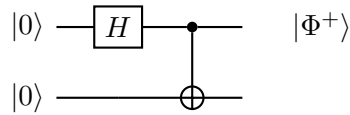
- In $|\Phi^+\rangle$ and $|\Phi^-\rangle$, the two qubits have **identical outcomes** upon measurement in the computational basis.
- In $|\Psi^+\rangle$ and $|\Psi^-\rangle$, the two qubits have **opposite outcomes**.
- The relative $\pm$ sign encodes an overall phase (interference between terms).

Generation from separable states. We can create these states from $|00\rangle$ by local unitary operations:

$$|\Phi^+\rangle = (H \otimes I) \text{CNOT} |00\rangle.$$

Other Bell states follow from simple single-qubit Pauli operations:

$$\begin{aligned} |\Phi^-\rangle &= (Z \otimes I) |\Phi^+\rangle, \\ |\Psi^+\rangle &= (X \otimes I) |\Phi^+\rangle, \\ |\Psi^-\rangle &= (XZ \otimes I) |\Phi^+\rangle. \end{aligned}$$



Measurement correlations. If Alice measures her qubit in $|\Phi^+\rangle$, Bob's qubit collapses correspondingly:

- If A measures 0, then B collapses to $|0\rangle$,
- If A measures 1, then B collapses to $|1\rangle$.

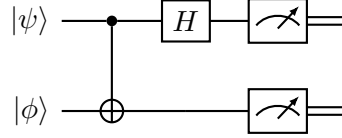
However, prior to measurement, neither qubit has a definite value—only the joint correlation exists.

3.3 Bell Measurement and Basis Transformation

A **Bell measurement** projects two qubits onto the Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$. Operationally, this measurement can be implemented by applying the inverse of the Bell-state generating circuit—namely, a CNOT followed by a Hadamard—then measuring in the computational basis.

$$\text{Bell measurement} = (H \otimes I)\text{CNOT}^\dagger.$$

Since the CNOT is its own inverse, this is just $(H \otimes I)\text{CNOT}$.



The two classical outcomes of measurement correspond to one of the four Bell states.

3.4 Quantum Teleportation

Teleportation uses an entangled pair and two classical bits to transfer an unknown qubit state from Alice to Bob without physically sending the qubit.

Step 1: Initial setup. Alice wants to send the state

$$|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle.$$

She and Bob share an entangled Bell pair

$$|\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The combined state is:

$$|\Psi\rangle_{123} = |\psi\rangle_1 \otimes |\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle).$$

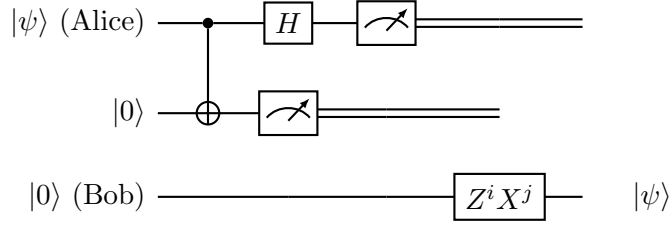
Step 2: Alice applies CNOT and Hadamard. Applying a CNOT from qubit 1 to 2 and then a Hadamard on qubit 1 yields:

$$\begin{aligned} |\Psi'\rangle_{123} &= (H_1 \otimes \text{CNOT}_{12})|\Psi\rangle_{123} \\ &= \frac{1}{2} \left[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) \right. \\ &\quad \left. + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle) \right]. \end{aligned}$$

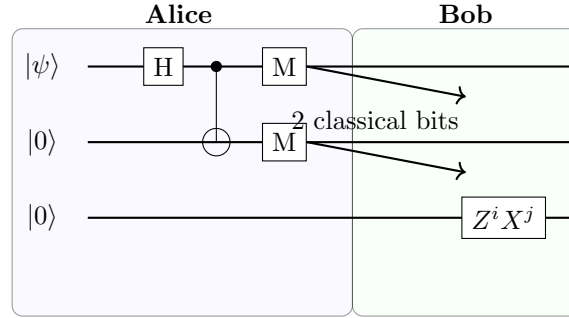
Step 3: Alice measures qubits 1 and 2. Measurement collapses the first two qubits into one of the four computational states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, leaving Bob's qubit (3) in one of the corresponding conditional states:

$$\begin{aligned} |00\rangle &\Rightarrow |\psi_3\rangle = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle, \\ |01\rangle &\Rightarrow X|\psi\rangle, \\ |10\rangle &\Rightarrow Z|\psi\rangle, \\ |11\rangle &\Rightarrow XZ|\psi\rangle. \end{aligned}$$

Step 4: Classical communication and correction. Alice sends her 2-bit outcome (i, j) to Bob. Bob applies $Z^i X^j$ to recover the original state $|\psi\rangle$ exactly.



3.5 Diagrammatic View



Quantum teleportation protocol: Alice performs local operations and sends classical bits to Bob, who recovers the state via local corrections.

3.6 The No-Cloning Theorem

Teleportation does not violate the impossibility of duplicating arbitrary quantum states. The **no-cloning theorem** states that there exists no unitary U satisfying:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

for all $|\psi\rangle$.

Proof sketch. Suppose it were possible. Then for $|\psi_1\rangle$ and $|\psi_2\rangle$,

$$\langle\psi_1|\psi_2\rangle = \langle\psi_1|\psi_2\rangle^2,$$

implying $\langle\psi_1|\psi_2\rangle = 0$ or 1 . But arbitrary states may have any overlap between 0 and 1 . Hence, cloning is impossible.

3.7 Interference and Quantum Algorithm Construction

Quantum systems exhibit **interference** when probability amplitudes combine coherently. For two indistinguishable alternatives with amplitudes ψ_1 and ψ_2 ,

$$P = |\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2|\psi_1||\psi_2|\cos\theta.$$

The phase difference θ controls constructive or destructive interference.

In circuits. Quantum gates such as H , S , and T introduce controlled phase changes, allowing quantum algorithms to manipulate interference patterns to amplify “correct” answers and suppress “incorrect” ones.

Example: Double-slit analogy. Just as waves passing through two slits interfere, qubit amplitudes interfere when multiple computational paths lead to the same outcome. Algorithms like Grover’s search deliberately tune phases to steer amplitude toward target states.

Intuitive steps in quantum algorithm design:

1. Initialize qubits to $|0\rangle^{\otimes n}$.
2. Apply Hadamards to create a superposition over all basis states.
3. Apply oracle or problem-specific unitaries to tag desired solutions with phase shifts.
4. Interfere amplitudes via further unitaries to amplify desired outcomes.
5. Measure to obtain a high-probability correct result.

3.8 Summary

- Entanglement creates correlations that have no classical analogue.
- The four Bell states form a maximally entangled basis for two-qubit systems.
- Quantum teleportation transfers an unknown qubit using entanglement and two classical bits.
- The no-cloning theorem ensures teleportation replaces, not copies, the original.
- Interference, when combined with entanglement, underlies the speedup in quantum algorithms.

Together, **superposition**, **entanglement**, and **interference** form the triad that makes quantum computation fundamentally distinct from classical computing.

4 Quantum Algorithms: Deutsch–Jozsa and Bernstein–Vazirani

4.1 Introduction and Motivation

Quantum algorithms exploit the unique principles of quantum mechanics—superposition, interference, and entanglement—to perform certain computational tasks more efficiently than any classical computer.

Among the earliest such algorithms are the **Deutsch–Jozsa (DJ)** and **Bernstein–Vazirani (BV)** algorithms. Although conceptually simple, they illustrate the essence of quantum speedup and serve as a foundation for understanding more complex algorithms such as Shor’s and Grover’s.

Historical background.

Algorithm	Year	Key Feature
Deutsch (single-bit version)	1985	First example of a quantum advantage
Deutsch–Jozsa	1992	Deterministic exponential speedup (exact separation)
Bernstein–Vazirani	1993	Probabilistic separation (BQP vs BPP)
Simon’s Algorithm	1994	Exponential quantum-classical gap
Shor’s Algorithm	1994	Polynomial-time factoring and period finding
Grover’s Algorithm	1996	Quadratic speedup for unstructured search

These algorithms were initially designed not for practical computation, but to demonstrate fundamental separations in computational **complexity classes** between classical and quantum models.

4.2 Oracle and Query Complexity Model

Many quantum algorithms are defined within the **oracle (black-box)** model. An oracle represents an unknown Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, accessible only via queries. The objective is to determine some property of f using as few oracle queries as possible.

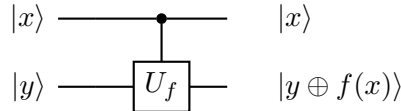
Classical oracle. The classical oracle simply returns the function value for a given input:

$$x \mapsto f(x).$$

Quantum oracle. The quantum oracle must be unitary. It acts on computational basis states as:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle,$$

where the second qubit stores the XOR (mod 2) of its previous value with $f(x)$.



The key insight is that the oracle acts on *superpositions*, effectively evaluating the function for all 2^n inputs simultaneously—a phenomenon known as **quantum parallelism**.

Query complexity. If an algorithm can solve a problem using fewer calls to the oracle than any classical algorithm, it demonstrates a **quantum speedup**. For example:

$$\text{Classical: } \mathcal{O}(2^{n-1} + 1) \quad \text{vs. Quantum: 1 query (Deutsch–Jozsa).}$$

4.3 The Deutsch–Jozsa Problem

We are given a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ guaranteed to be either:

- **Constant:** $f(x) = 0$ or 1 for all inputs.
- **Balanced:** $f(x) = 0$ for half of the inputs and 1 for the other half.

Our goal is to determine whether f is constant or balanced.

Classical complexity. A deterministic classical algorithm must query at least $2^{n-1} + 1$ distinct inputs in the worst case to guarantee the answer.

Quantum complexity. The Deutsch–Jozsa algorithm requires only **one query** to the oracle and produces the answer deterministically.

4.4 Deutsch–Jozsa Algorithm: Step-by-Step Derivation

Step 0: Initialization. Start with $n + 1$ qubits:

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle.$$

Step 1: Apply Hadamards. Apply Hadamard gates to all qubits:

$$|\psi_1\rangle = (H^{\otimes n} \otimes H) |0\rangle^{\otimes n} |1\rangle.$$

Using $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, we get:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle (|0\rangle - |1\rangle).$$

Step 2: Apply the oracle.

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

For the superposition $(|0\rangle - |1\rangle)/\sqrt{2}$, we obtain a **phase kickback**:

$$U_f \left(|x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Hence,

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The second register can now be ignored since it no longer affects the first.

Step 3: Apply Hadamard to the first n qubits. For each basis state $|x\rangle$:

$$H^{\otimes n} |x\rangle = \frac{1}{2^{n/2}} \sum_z (-1)^{x \cdot z} |z\rangle,$$

where $x \cdot z$ is the bitwise inner product (mod 2). Applying this transform:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_z \left[\sum_x (-1)^{f(x) + x \cdot z} \right] |z\rangle.$$

Step 4: Measure the first n qubits. The amplitude for $|z\rangle$ is:

$$A_z = \frac{1}{2^n} \sum_x (-1)^{f(x) + x \cdot z}.$$

Case 1: Constant function. If $f(x)$ is constant, then:

$$A_0 = \pm 1, \quad A_{z \neq 0} = 0.$$

Measurement always yields $|0\rangle^{\otimes n}$.

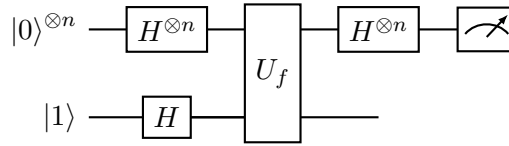
Case 2: Balanced function. If $f(x)$ is balanced, half of the terms in the sum cancel:

$$A_0 = 0.$$

Therefore, the probability of observing $|0\rangle^{\otimes n}$ is zero.

Result.

$$\begin{cases} \text{Measure } |0\rangle^{\otimes n} \Rightarrow f \text{ is constant,} \\ \text{Measure any other state} \Rightarrow f \text{ is balanced.} \end{cases}$$



Conclusion. The Deutsch–Jozsa algorithm solves the problem in a single query, deterministically, while any classical deterministic algorithm would require exponentially many queries.

4.5 Understanding Phase Kickback

The heart of many quantum algorithms lies in **phase kickback**. When the oracle applies a conditional phase on a target qubit in the state $(|0\rangle - |1\rangle)/\sqrt{2}$, the function value is “kicked back” as a phase on the control qubits:

$$U_f \left(|x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

This phenomenon allows quantum algorithms to encode information in phase rather than amplitude, which is crucial for interference-based speedups.

4.6 Bernstein–Vazirani Problem

The BV algorithm generalizes Deutsch–Jozsa to find an unknown bit string $a \in \{0, 1\}^n$ hidden inside a linear Boolean function:

$$f_a(x) = a \cdot x = a_1 x_1 \oplus a_2 x_2 \oplus \cdots \oplus a_n x_n.$$

Goal. Determine a using the fewest possible oracle queries.

Classical approach. To determine all bits of a , a classical algorithm must query f on each of the n unit vectors e_i , requiring n queries.

Quantum approach. The BV algorithm recovers the entire bit string a with a single quantum query.

4.7 Bernstein–Vazirani Algorithm Derivation

Step 0: Initialization.

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle.$$

Step 1: Apply Hadamards.

$$|\psi_1\rangle = (H^{\otimes n} \otimes H) |0\rangle^{\otimes n} |1\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_x |x\rangle (|0\rangle - |1\rangle).$$

Step 2: Apply oracle U_{f_a} .

$$U_{f_a} |x\rangle |y\rangle = |x\rangle |y \oplus (a \cdot x)\rangle.$$

By phase kickback,

$$U_{f_a} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} = (-1)^{a \cdot x} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Thus,

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{a \cdot x} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Step 3: Apply Hadamards to the first n qubits.

$$H^{\otimes n} |x\rangle = \frac{1}{2^{n/2}} \sum_z (-1)^{x \cdot z} |z\rangle,$$

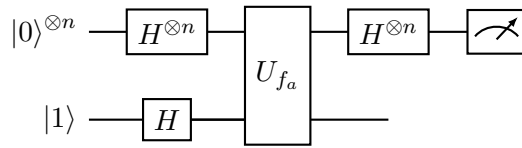
leading to:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_z \left[\sum_x (-1)^{a \cdot x + x \cdot z} \right] |z\rangle.$$

The inner sum evaluates to 2^n when $z = a$ and 0 otherwise, giving:

$$|\psi_3\rangle = |a\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Step 4: Measurement. Measuring the first n qubits yields $|a\rangle$ with certainty.



Interpretation. The BV algorithm performs all n evaluations of f_a simultaneously in superposition, encoding the complete string a as a phase pattern. A final Hadamard transformation extracts a in one measurement.

4.8 Complexity-Theoretic Perspective

- **Deutsch–Jozsa:** Separates EQP (Exact Quantum Polynomial time) from P (deterministic classical polynomial time).
- **Bernstein–Vazirani:** Illustrates the power of BQP (Bounded-error Quantum Polynomial time) over BPP (Bounded-error Probabilistic Polynomial time).

$$\mathbf{P} \subsetneq \mathbf{BPP} \subseteq \mathbf{BQP}.$$

While these problems are of limited practical relevance, they demonstrate the essential mechanisms behind all quantum speedups:

1. **Superposition:** enables parallel exploration of all inputs.
2. **Phase kickback:** encodes function information without measurement.
3. **Interference:** amplifies correct solutions while cancelling others.

4.9 Summary

- The Deutsch–Jozsa algorithm distinguishes constant vs balanced Boolean functions with one query.
- The Bernstein–Vazirani algorithm determines a hidden n -bit string in one query.
- Both algorithms leverage the principles of quantum parallelism and interference.
- These algorithms laid the conceptual groundwork for later breakthroughs such as Simon’s and Shor’s algorithms.

Key Takeaway: Deutsch–Jozsa and Bernstein–Vazirani show that quantum mechanics allows global properties of functions to be extracted with exponentially fewer oracle calls than classical computation—demonstrating the first formal evidence of quantum advantage.

5 Grover’s Search Algorithm: Amplitude Amplification and Quantum Search

5.1 Motivation and Overview

Grover’s algorithm, proposed by Lov Grover in 1996, is one of the most celebrated quantum algorithms. It provides a **quadratic speedup** for unstructured search problems, reducing the number of oracle queries from $O(N)$ in classical computation to $O(\sqrt{N})$ in the quantum regime.

Although this is not an exponential speedup, it demonstrates the power of quantum interference and superposition in amplifying the probability amplitude of desired outcomes. It is also the foundational prototype of a more general technique known as **amplitude amplification**.

Problem Statement. Let $N = 2^n$ be the number of possible states of an n -qubit system. We are given a Boolean oracle $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that:

$$f(x) = \begin{cases} 1, & \text{if } x = x^* \text{ (the marked item),} \\ 0, & \text{otherwise.} \end{cases}$$

Our task is to find the marked item x^* .

A classical search must check each element sequentially, requiring on average $N/2$ queries. Grover’s algorithm can locate x^* with high probability using approximately $\frac{\pi}{4}\sqrt{N}$ queries.

5.2 Oracle and Phase Kickback Mechanism

The quantum oracle is implemented as a unitary operator O_f :

$$O_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

If we prepare the second (target) qubit in the state

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

then:

$$O_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

Thus, the oracle flips the phase of the marked state(s):

$$O_f |x\rangle = \begin{cases} -|x\rangle, & \text{if } f(x) = 1, \\ |x\rangle, & \text{if } f(x) = 0. \end{cases}$$

This **phase inversion** is the key step—rather than storing the function value in an ancillary qubit, the information about $f(x)$ is encoded as a relative phase, which can interfere in subsequent steps.

5.3 Initialization: Uniform Superposition

We initialize all n qubits in the $|0\rangle$ state:

$$|\psi_0\rangle = |0\rangle^{\otimes n}.$$

Applying the Hadamard gate to each qubit yields a uniform superposition over all possible basis states:

$$|\psi_0\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

Each computational basis state $|x\rangle$ now has amplitude $1/\sqrt{N}$.

At this stage, no measurement would reveal useful information; the system's state represents maximal uncertainty about which item is marked.

5.4 Two-Dimensional Subspace Representation

Despite the high-dimensional Hilbert space (2^n), Grover's algorithm evolves entirely within a *two-dimensional subspace* spanned by:

$$|a\rangle = |x^*\rangle, \quad |e\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq x^*} |x\rangle.$$

These two orthonormal states represent: - $|a\rangle$: the **marked state**, and - $|e\rangle$: the **equal superposition** of all unmarked states.

We can decompose the initial uniform superposition as:

$$|\psi_0\rangle = \sin(\theta_0) |a\rangle + \cos(\theta_0) |e\rangle ,$$

where the angle θ_0 satisfies:

$$\sin(\theta_0) = \frac{1}{\sqrt{N}}, \quad \cos(\theta_0) = \sqrt{\frac{N-1}{N}}.$$

—

5.5 Geometric Intuition: Reflections as Rotations

The essence of Grover's algorithm is geometric. Each iteration performs a rotation in the two-dimensional plane spanned by $\{|a\rangle, |e\rangle\}$, bringing the state vector closer to $|a\rangle$.

The algorithm achieves this through two reflections:

$$G = D O_f,$$

where O_f and D are both unitary and Hermitian.

(1) Oracle reflection:

$$O_f = I - 2 |a\rangle \langle a| .$$

This operator reflects the current state vector about the hyperplane orthogonal to $|a\rangle$, introducing a negative sign to the amplitude of the marked state.

(2) Diffusion operator (Inversion about the mean):

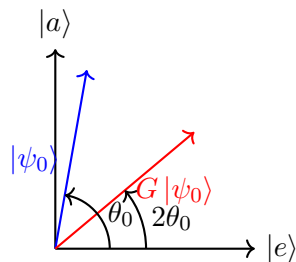
$$D = 2 |U\rangle \langle U| - I,$$

where $|U\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$ is the uniform superposition state.

The diffusion operator reflects the quantum state about the vector $|U\rangle$, effectively “inverting” the amplitudes about their mean value.

Combined effect: Two reflections about intersecting axes in a plane produce a rotation by twice the angle between them. Thus, $G = D O_f$ performs a rotation by $2\theta_0$ toward the target state $|a\rangle$.

—



Each Grover iteration increases the amplitude of the marked state and decreases that of all others.

5.6 Algebraic Evolution of the State

Let the state after t iterations be:

$$|\psi_t\rangle = \sin((2t+1)\theta_0) |a\rangle + \cos((2t+1)\theta_0) |e\rangle.$$

The oracle O_f first flips the sign of the amplitude associated with $|a\rangle$, and the diffusion operator D reflects the resulting state about $|U\rangle$. Together, they additively increase the projection onto $|a\rangle$ by $2\theta_0$ each iteration.

Example (small N): For $N = 4$, $\sin \theta_0 = 1/2 \Rightarrow \theta_0 = \pi/6$. Thus, after one iteration,

$$(2t+1)\theta_0 = 3\theta_0 = \pi/2,$$

and the state aligns exactly with $|a\rangle$ —a perfect search in one query.

5.7 Amplitude Amplification and Optimal Iteration Count

After t Grover iterations:

$$P_t = |\langle a|\psi_t\rangle|^2 = \sin^2((2t+1)\theta_0).$$

To maximize this probability, we choose:

$$(2t+1)\theta_0 = \frac{\pi}{2} \Rightarrow t_{\text{opt}} \approx \frac{\pi}{4\theta_0} - \frac{1}{2}.$$

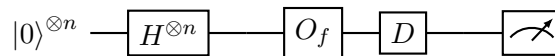
Since $\sin \theta_0 = 1/\sqrt{N}$, for large N , $\theta_0 \approx 1/\sqrt{N}$, giving:

$$t_{\text{opt}} = O(\sqrt{N}).$$

After this number of iterations, the measurement yields the marked item with probability close to 1.

Over-rotation. If additional iterations are applied beyond the optimum, the amplitude begins to rotate past the marked state, reducing success probability. Thus, precise control of the iteration count is essential.

5.8 Circuit Implementation



Steps:

1. Apply $H^{\otimes n}$ to prepare the uniform superposition.

2. Apply the oracle O_f to mark the target state via phase inversion.
3. Apply the diffusion operator D to amplify the marked amplitude.
4. Repeat the oracle–diffusion sequence t_{opt} times.
5. Measure in the computational basis.

Diffusion operator implementation. The operator D can be decomposed as:

$$D = H^{\otimes n} (2|0\rangle\langle 0| - I) H^{\otimes n}.$$

Hence, in circuit form: 1. Apply Hadamards, 2. Apply a conditional phase flip on $|0\rangle^{\otimes n}$, 3. Apply Hadamards again.

This construction requires only standard gates (H , Z , and multi-controlled Z).

5.9 Mathematical Proof of Rotation Behavior

Let $|\psi_t\rangle$ and $|\psi_{t+1}\rangle$ denote consecutive states. Each iteration satisfies:

$$|\psi_{t+1}\rangle = G|\psi_t\rangle = (2|U\rangle\langle U| - I)(I - 2|a\rangle\langle a|)|\psi_t\rangle.$$

Expanding in the $\{|a\rangle, |e\rangle\}$ basis yields:

$$G = \begin{pmatrix} \cos(2\theta_0) & -\sin(2\theta_0) \\ \sin(2\theta_0) & \cos(2\theta_0) \end{pmatrix},$$

showing that G acts as a rotation by angle $2\theta_0$ in the two-dimensional plane. Thus, the amplitude of the marked state grows linearly with iteration count, while the success probability grows quadratically until saturation.

5.10 Algorithm Summary

Grover's Algorithm: Step-by-Step

1. Initialize: $|0\rangle^{\otimes n}$.
 2. Apply Hadamard gates to obtain the uniform superposition: $|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$.
 3. Repeat for $t = \lfloor \frac{\pi}{4 \arcsin(1/\sqrt{N})} \rfloor$:
 - (a) Apply the phase oracle O_f .
 - (b) Apply the diffusion operator $D = 2|U\rangle\langle U| - I$.
 4. Measure in the computational basis. The result equals the marked state x^* with probability $> 1 - O(1/N)$.
-

5.11 Complexity and Optimality

- **Time complexity:** $O(\sqrt{N})$ oracle calls.
- **Space complexity:** $O(\log N)$ qubits.
- **Optimality:** Proven by Bennett, Bernstein, Brassard, and Vazirani (1997) that no quantum algorithm can find a marked element in an unstructured database with fewer than $O(\sqrt{N})$ queries.

Classical: $O(N)$	Quantum: $O(\sqrt{N})$
-------------------	------------------------

5.12 Extensions and Generalizations

- **Multiple marked items:** If k solutions exist, the optimal iteration count becomes $t_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{k}}$.
- **Amplitude amplification:** Grover's procedure is a special case of a broader framework that can amplify the success probability of any quantum algorithm.
- **Unknown k :** Adaptive or quantum counting techniques can estimate k to within a known precision.

5.13 Summary and Conceptual Insights

Grover's search demonstrates that quantum computation can exploit interference to amplify the probability of correct answers. The two-step process—phase inversion followed by inversion about the mean—rotates the system's state vector toward the desired solution in Hilbert space.

Physical intuition:

- The oracle introduces a relative phase that encodes which state is marked.
- The diffusion operator redistributes amplitude globally, increasing the marked amplitude.
- Each iteration acts as a geometric rotation, enhancing constructive interference.

Grover's algorithm is thus a perfect illustration of how quantum mechanics replaces classical probability with complex amplitude manipulation—*search by interference rather than enumeration*.

6 Variational Quantum Algorithms and the NISQ Era

6.1 Motivation

The current generation of quantum hardware, characterized by devices with tens to a few hundred noisy qubits, is often referred to as the **Noisy Intermediate-Scale Quantum (NISQ)** era [1].

These devices are not yet fault-tolerant and cannot execute deep quantum circuits reliably due to noise and decoherence. Nevertheless, they offer computational capabilities that may surpass classical systems for certain tasks.

Traditional quantum algorithms such as Shor’s and Grover’s require long coherent evolutions and perfect gates. In contrast, **Variational Quantum Algorithms (VQAs)** are designed to function within the constraints of NISQ devices. They employ a **hybrid quantum–classical loop**, where the quantum processor prepares parameterized states and measures expectation values, while a classical optimizer updates the parameters to minimize a cost function.

6.2 Structure of a VQA

A general VQA consists of:

1. **Ansatz preparation:** The quantum circuit prepares a parameterized state

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle^{\otimes n},$$

where $U(\boldsymbol{\theta})$ is a unitary composed of rotation and entangling gates.

2. **Cost function:** A real-valued function $C(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle$ representing an observable (e.g., the energy of a Hamiltonian).
3. **Classical optimization:** A classical algorithm updates parameters $\boldsymbol{\theta}$ iteratively to minimize $C(\boldsymbol{\theta})$.

This hybrid structure allows shallow circuits, making VQAs resilient to noise.

6.3 Representative Variational Algorithms

- **Variational Quantum Eigensolver (VQE):** Estimates the ground state energy of a molecular Hamiltonian H using the variational principle

$$E_0 \leq \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle.$$

Widely applied in quantum chemistry and materials science.

- **Quantum Approximate Optimization Algorithm (QAOA):** Solves combinatorial optimization problems by alternating between problem and mixer Hamiltonians.
- **Quantum Generative Adversarial Networks (qGANs):** Combine a quantum generator with a classical discriminator for sampling and learning probability distributions (e.g., option pricing in finance).

6.4 Challenges and Mitigation

Despite their potential, VQAs face several limitations:

- **Barren plateaus:** Flat cost landscapes cause vanishing gradients, hindering trainability.
- **Noise sensitivity:** Decoherence and gate errors distort measured expectation values.

- **Measurement overhead:** Estimating many expectation values demands a large number of circuit executions.

Mitigation strategies include using problem-inspired ansatzes, pre-training techniques, adaptive optimizers such as Simultaneous Perturbation Stochastic Approximation (SPSA), and error mitigation techniques like **zero-noise extrapolation** and **subspace expansion**.

6.5 Applications and Outlook

VQAs have demonstrated applicability in a wide range of domains:

- **Quantum Chemistry:** Ground-state energy estimation and molecular simulation (VQE).
- **Finance:** Option pricing and portfolio optimization (qGANs, QAOA).
- **Optimization:** Supply chain and logistics problems.
- **Machine Learning:** Quantum classifiers and variational neural networks.

Programmable quantum computing is a reality, but scalability remains limited by noise and decoherence. Hybrid variational frameworks are expected to dominate the NISQ era and may serve as a bridge toward future fault-tolerant quantum computing.

7 Quantum Error Correction (QEC)

7.1 Motivation: From Decoherence to Error Correction

Quantum information is encoded in fragile superpositions that are highly susceptible to noise arising from unwanted interactions with the environment (bath). This phenomenon, known as **decoherence**, leads to the irreversible loss of phase information and results in mixed states. A scalable quantum computer must therefore fulfill the following desiderata:

1. A scalable system with well-defined qubits,
2. Initialization to a simple fiducial state,
3. Long coherence times,
4. A universal set of quantum gates,
5. Reliable single-qubit measurements.

While small-scale demonstrations of algorithms (e.g., Grover’s search) succeed on ideal simulators, their performance degrades severely on real hardware due to noise and gate errors. John Preskill’s 2018 essay [1] emphasized that Quantum Error Correction (QEC) is the foundation for scalable quantum computation, allowing noisy physical qubits to simulate reliable logical qubits.

7.2 Noise on Classical Bits: The 3-bit Repetition Code

Before understanding QEC, it is instructive to recall its classical counterpart. In a **binary symmetric channel**, a bit flips with probability p . A simple 3-bit repetition code encodes logical 0 and 1 as

$$0 \mapsto 000, \quad 1 \mapsto 111.$$

If a single bit flips (e.g., $000 \rightarrow 010$ or $111 \rightarrow 101$), majority voting allows error detection and correction. The decoder computes parity checks and restores the most likely original codeword. This code corrects single-bit errors and suppresses first-order error probability to $\mathcal{O}(p^2)$.

7.3 Noise in Quantum Systems

In quantum systems, noise is more complex since quantum evolution may not remain unitary. For instance, under a **bit-flip channel**, a qubit in the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

undergoes a random Pauli- X error with probability p :

$$\rho \mapsto (1-p)\rho + pX\rho X.$$

For a pure input $|0\rangle$, the output becomes a mixed state

$$\rho = (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|.$$

The error probability behaves as

$$p = \frac{1}{2}(1 - e^{-t/T}),$$

where T denotes the coherence time of the qubit.

7.4 Challenges in Quantum Error Correction

Unlike classical bits, quantum information cannot be directly copied due to the **no-cloning theorem**. Furthermore:

1. Quantum errors form a *continuum* of possibilities (bit-flip, phase-flip, amplitude damping, etc.).
2. Measurement disturbs the state and can destroy coherence.

QEC must therefore overcome these challenges through the following ideas:

- **Entanglement:** Encode logical information into entangled codewords spread across multiple qubits.
- **Error discretization:** Represent general errors as linear combinations of a finite unitary basis $\{I, X, Y, Z\}$.
- **Syndrome extraction:** Detect errors using controlled gates and ancilla qubits without collapsing the logical state.

7.5 The 3-Qubit Bit-Flip Code

The simplest quantum code that protects against single bit-flip errors is the 3-qubit repetition (bit-flip) code. It encodes one logical qubit into three physical qubits:

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle.$$

A general logical state is

$$|\psi_L\rangle = \alpha |0_L\rangle + \beta |1_L\rangle = \alpha |000\rangle + \beta |111\rangle.$$

The no-cloning theorem forbids copying the unknown state $\alpha |0\rangle + \beta |1\rangle$ directly, but we can entangle it with two auxiliary qubits initially in $|0\rangle$ using CNOT operations.

7.6 Encoding Circuit

The encoding circuit uses two CNOT gates. After encoding, noise acts independently on each physical qubit. The dominant single-qubit error operators are

$$XII, \quad IXI, \quad IIX,$$

each occurring with probability p . Multi-qubit errors are of higher order $\mathcal{O}(p^2)$ or $\mathcal{O}(p^3)$.

7.7 Effect of Errors on Codewords

If one physical qubit flips, the codewords transform as

$$\begin{aligned} |000\rangle &\rightarrow |100\rangle, |010\rangle, |001\rangle, \\ |111\rangle &\rightarrow |011\rangle, |101\rangle, |110\rangle. \end{aligned}$$

Each corrupted state is orthogonal to the others, which makes different error patterns distinguishable through appropriate measurements.

7.8 Syndrome Measurement and Error Recovery

Error detection is performed indirectly using two ancillary qubits, which measure the parity between neighboring physical qubits in the Z basis:

$$Z_1 Z_2, \quad Z_2 Z_3.$$

The outcomes of these measurements, called **syndrome bits**, identify which qubit (if any) has flipped. A typical syndrome table is:

Error	$Z_1 Z_2$	$Z_2 Z_3$	Correction
I (no error)	+1	+1	–
X_1	-1	+1	Apply X_1
X_2	-1	-1	Apply X_2
X_3	+1	-1	Apply X_3

The recovery operation applies a Pauli- X gate to the identified qubit, restoring the original logical state. Hence, this code *perfectly corrects all single-qubit bit-flip errors*.

7.9 Performance and Limitations

The 3-qubit code improves the fidelity of the logical qubit, provided the error rate $p < \frac{1}{2}$. It cannot, however, correct double or triple errors. To protect against both bit- and phase-flip errors, one needs larger codes (e.g., the 5-qubit or 7-qubit Steane code). The general framework of QEC uses stabilizer formalism and the Pauli group to describe syndrome extraction and recovery.

7.10 Resource Requirements and Fault-Tolerance

Real quantum systems experience various error processes—erasure, phase-flip (T_2), amplitude-damping (T_1), and depolarization. By discretizing these errors into $\{I, X, Y, Z\}$, QEC reduces a continuum of noise to a finite set of correctable errors. The **Quantum Hamming Bound** states that to correct arbitrary single-qubit errors perfectly, at least five physical qubits are required:

$$2^k(3n + 1) \leq 2^n.$$

Further reliability is achieved through:

- **Fault-tolerant circuits** — to ensure that errors do not propagate uncontrollably,
- **Concatenation** — encoding logical qubits within larger codes (e.g., a concatenated 7-qubit code uses $7^2 = 49$ qubits per logical qubit).

7.11 Outlook

Quantum error correction lies at the core of scalable quantum computing. It enables the construction of logical qubits with arbitrarily low effective error rates when combined with fault-tolerant architectures. Recent industrial roadmaps (IBM 2021, Google 2018) envision logical qubits supported by surface codes operating over thousands of physical qubits. As succinctly expressed by Daniel Gottesman’s “QEC Sonnet”:

*With group and eigenstate, we’ve learned to fix,
Your quantum errors with our quantum tricks.*

References

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